



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECP3623

Database Programming

2025/2026 SEMESTER 1

Assignment 1 (10%)

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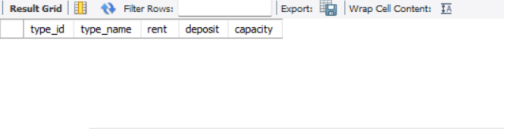
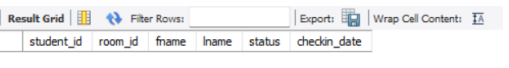
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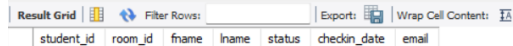
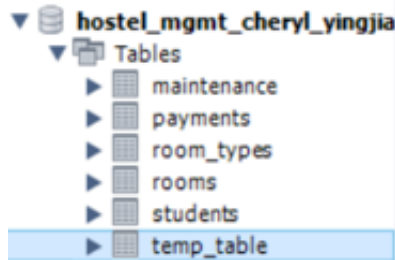
Task	Criteria	Marks	Total Marks
Task 1 Database Creation & Constraints	Creation of database and all tables with valid keys, datatypes, and constraints (PK, FK, NOT NULL, UNIQUE). Logical order and referential integrity preserved.	3	
	ALTER TABLE (add email UNIQUE) and DROP TABLE demonstration executed successfully and documented.	2	
	Insertion order, valid FK references, and no orphan data.	2	
Task 2 Data Manipulation & Filtering	Five tables populated with ≥10 valid records each; varied and meaningful data UPDATE/DELETE executed correctly with logical results.	1	
	Accurate use of WHERE, BETWEEN, LIKE, IN, AND/OR/NOT queries.	3	
	Correct use of aggregate and string functions (COUNT, CONCAT, and more) .	2	
Task 3 Reporting & Aggregation	VIEW creation	2	
	Use of GROUP BY, HAVING, ROUND, and CASE with accurate aggregated results.	3	
	Clear labelling, readable outputs, screenshots show executed results.	2	
Total (20)			

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Task 3 – Reporting and Aggregation (CLO2)	10

Task 1-Database Creation and Integrity Constraints (CLO1)

	Code	Output
1.	<pre>CREATE DATABASE hostel_mgmt_Cheryl_YingJia;</pre> Create a database named hostel_mgmt_Cheryl_YingJia	
2.	<pre>CREATE TABLE room_types (type_id int AUTO_INCREMENT PRIMARY KEY, type_name VARCHAR (255) NOT NULL, rent DECIMAL (10,2) NOT NULL, deposit DECIMAL (10,2) NOT NULL, capacity INT NOT NULL);</pre> a. Create table named room_types with the entity type_id, type_name, rent, deposit and capacity	
	<pre>CREATE TABLE rooms (room_id int AUTO_INCREMENT PRIMARY KEY, type_id int, room_no VARCHAR (50) NOT NULL, floor_no int NOT NULL, is_occupied BOOLEAN NOT NULL, FOREIGN KEY (type_id) REFERENCES room_types(type_id));</pre> b. Create table named rooms with the entity room_id, type_id, room_no, floor_no and is_occupied. The attribute type_id is used as a foreign key to reference the room_types table.	
	<pre>CREATE TABLE students (student_id int AUTO_INCREMENT PRIMARY KEY, room_id int NOT NULL, fname VARCHAR (255) NOT NULL, lname VARCHAR (255) NOT NULL, status ENUM('ACTIVE', 'NON_ACTIVE') NOT NULL, checkin_date DATE, FOREIGN KEY (room_id) REFERENCES rooms (room_id));</pre> c. Create table named students with the entity student_id, room_id, fname, lname, status and checkin_date. The attribute room_id is used as a foreign key to reference the rooms table.	
	<pre>CREATE TABLE maintenance (maint_id int AUTO_INCREMENT PRIMARY KEY, room_id int NOT NULL, issue_desc TEXT NOT NULL, severity ENUM ('LOW', 'MEDIUM', 'HIGH') NOT NULL, status ENUM ('OPEN', 'RESOLVED') NOT NULL, reported_on DATE NOT NULL, resolved_on DATE NULL, FOREIGN KEY (room_id) REFERENCES rooms (room_id));</pre> d.	

	Create table named maintenance with the entity maint_id, room_id, issue_desc, severity, status, reported_on and resolved_on. The attribute room_id is used as a foreign key to reference the rooms table.	
	<pre> CREATE TABLE payments (payment_id int AUTO_INCREMENT PRIMARY KEY, student_id int NOT NULL, amount DECIMAL (10,2), paid_on DATE NOT NULL, method ENUM ('CASH','FPX','CARD','TNG') NOT NULL, note VARCHAR (255), FOREIGN KEY (student_id) REFERENCES students (student_id)); </pre> e. Create table named payments with the entity payment_id, student_id, amount, paid_on, method and note. The attribute student_id is used as a foreign key to reference the students table.	e. 
4.	<pre> UPDATE rooms r SET is_occupied = EXISTS (SELECT 1 FROM students s WHERE s.room_id = r.room_id AND s.status = 'ACTIVE') WHERE r.room_id > 0; </pre> A room is 'occupied' if it has ≥ 1 student with status='ACTIVE'; otherwise 'VACANT'	
5.	<pre> ALTER TABLE students ADD email VARCHAR (255) UNIQUE; </pre> Use ALTER TABLE to add a new column email to the students table and mark it as UNIQUE	
	<pre> CREATE TABLE temp_table (temp_id int); DROP TABLE IF EXISTS temp_table; </pre> Create a temporary table named temp_table and remove it safely using DROP TABLE IF EXISTS	

		▼  hostel_mgmt_cheryl_yingjia ▼  Tables ▶  maintenance ▶  payments ▶  room_types ▶  rooms ▶  students ▼  Views
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Task 2 - Data Manipulation and Filtering (CLO2)

	Code	Output																																																																																																									
1.	<pre>INSERT INTO room_types(type_name, rent, deposit, capacity) VALUES ('Single Standard', 300.00, 150.00, 1), ('Double Standard', 400.00, 200.00, 2), ('Single Deluxe', 400.00, 200.00, 1), ('Double Deluxe', 500.00, 250.00, 2), ('Single Premium', 500.00, 250.00, 1), ('Double Premium', 600.00, 300.00, 2), ('Family Standard (Small)', 900.00, 500.00, 4), ('Family Standard (Large)', 1200.00, 600.00, 6), ('Family Deluxe (Small)', 1100.00, 300.00, 1), ('Family Deluxe (Large)', 1400.00, 250.00, 2);</pre> Insert 10 data into room_types table. Different room types for single, double and family rooms. We have set the type_id to auto increment.	<table><tr><th></th><th>type_id</th><th>type_name</th><th>rent</th><th>deposit</th><th>capacity</th></tr><tr><td>▶</td><td>1</td><td>Single Standard</td><td>300.00</td><td>150.00</td><td>1</td></tr><tr><td></td><td>2</td><td>Double Standard</td><td>400.00</td><td>200.00</td><td>2</td></tr><tr><td></td><td>3</td><td>Single Deluxe</td><td>400.00</td><td>200.00</td><td>1</td></tr><tr><td></td><td>4</td><td>Double Deluxe</td><td>500.00</td><td>250.00</td><td>2</td></tr><tr><td></td><td>5</td><td>Single Premium</td><td>500.00</td><td>250.00</td><td>1</td></tr><tr><td></td><td>6</td><td>Double Premium</td><td>600.00</td><td>300.00</td><td>2</td></tr><tr><td></td><td>7</td><td>Family Standard (Small)</td><td>900.00</td><td>500.00</td><td>4</td></tr><tr><td></td><td>8</td><td>Family Standard (Large)</td><td>1200.00</td><td>600.00</td><td>6</td></tr><tr><td></td><td>9</td><td>Family Deluxe (Small)</td><td>1100.00</td><td>300.00</td><td>1</td></tr><tr><td></td><td>10</td><td>Family Deluxe (Large)</td><td>1400.00</td><td>250.00</td><td>2</td></tr></table>		type_id	type_name	rent	deposit	capacity	▶	1	Single Standard	300.00	150.00	1		2	Double Standard	400.00	200.00	2		3	Single Deluxe	400.00	200.00	1		4	Double Deluxe	500.00	250.00	2		5	Single Premium	500.00	250.00	1		6	Double Premium	600.00	300.00	2		7	Family Standard (Small)	900.00	500.00	4		8	Family Standard (Large)	1200.00	600.00	6		9	Family Deluxe (Small)	1100.00	300.00	1		10	Family Deluxe (Large)	1400.00	250.00	2																																							
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	<pre>INSERT INTO rooms (type_id, room_no, floor_no, is_occupied) VALUES (1, 'A101', 1, FALSE), (2, 'B111', 1, FALSE), (3, 'A201', 2, FALSE), (4, 'B211', 2, FALSE), (5, 'A301', 3, FALSE), (6, 'B311', 3, FALSE), (7, 'C401', 4, FALSE), (8, 'C411', 4, FALSE), (9, 'C501', 5, FALSE), (10, 'C512', 5, FALSE);</pre> Insert 10 data into rooms table. We follow the sequence for type_id then we assign room number, floor number and occupant of rooms.	<table><tr><th></th><th>room_id</th><th>type_id</th><th>room_no</th><th>floor_no</th><th>is_occupied</th></tr><tr><td>▶</td><td>1</td><td>1</td><td>A101</td><td>1</td><td>0</td></tr><tr><td></td><td>2</td><td>2</td><td>B111</td><td>1</td><td>0</td></tr><tr><td></td><td>3</td><td>3</td><td>A201</td><td>2</td><td>0</td></tr><tr><td></td><td>4</td><td>4</td><td>B211</td><td>2</td><td>0</td></tr><tr><td></td><td>5</td><td>5</td><td>A301</td><td>3</td><td>0</td></tr><tr><td></td><td>6</td><td>6</td><td>B311</td><td>3</td><td>0</td></tr><tr><td></td><td>7</td><td>7</td><td>C401</td><td>4</td><td>0</td></tr><tr><td></td><td>8</td><td>8</td><td>C411</td><td>4</td><td>0</td></tr><tr><td></td><td>9</td><td>9</td><td>C501</td><td>5</td><td>0</td></tr><tr><td></td><td>10</td><td>10</td><td>C512</td><td>5</td><td>0</td></tr></table>		room_id	type_id	room_no	floor_no	is_occupied	▶	1	1	A101	1	0		2	2	B111	1	0		3	3	A201	2	0		4	4	B211	2	0		5	5	A301	3	0		6	6	B311	3	0		7	7	C401	4	0		8	8	C411	4	0		9	9	C501	5	0		10	10	C512	5	0																																							
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	<pre>INSERT INTO students (room_id, fname, lname, status, checkin_date, email) VALUES (1, 'Ahmad', 'Mohammed', 'ACTIVE', '2025-08-02', 'ahmadm@graduate.utm.my'), (2, 'Yee Wen', 'Lau', 'ACTIVE', '2025-09-01', 'yeewen@graduate.utm.my'), (3, 'Cheryl', 'Cheong', 'ACTIVE', '2025-08-15', 'cheryl@graduate.utm.my'), (4, 'Dora', 'Poh', 'ACTIVE', '2025-09-05', 'pohdora@graduate.utm.my'), (5, 'Emily', 'Liew', 'ACTIVE', '2025-08-31', 'emily_l@graduate.utm.my'), (6, 'Jeffery', 'Gan', 'NON_ACTIVE', '2024-09-22', 'jeff@graduate.utm.my'), (7, 'Ganesh', 'Kumar', 'ACTIVE', '2025-10-27', 'ganesh_k@graduate.utm.my'), (8, 'Wyness', 'Ong', 'ACTIVE', '2025-08-17', 'wyness@graduate.utm.my'), (9, 'Ibrahim', 'Ali', 'NON_ACTIVE', '2024-10-23', 'ibrahimali@graduate.utm.my'), (10, 'Jasmine', 'Lee', 'ACTIVE', '2025-09-10', 'jasmine@graduate.utm.my');</pre> Insert 10 data into students table. We assigned students into rooms accordingly also with their name, status, check-in dates and their email.	<table><tr><th>student_id</th><th>room_id</th><th>fname</th><th>lname</th><th>status</th><th>checkin_date</th></tr><tr><td>1</td><td>1</td><td>Ahmad</td><td>Mohammed</td><td>ACTIVE</td><td>2025-08-02</td></tr><tr><td>2</td><td>2</td><td>Yee Wen</td><td>Lau</td><td>ACTIVE</td><td>2025-09-01</td></tr><tr><td>3</td><td>3</td><td>Cheryl</td><td>Cheong</td><td>ACTIVE</td><td>2025-08-15</td></tr><tr><td>4</td><td>4</td><td>Dora</td><td>Poh</td><td>ACTIVE</td><td>2025-09-05</td></tr><tr><td>5</td><td>5</td><td>Emily</td><td>Liew</td><td>ACTIVE</td><td>2025-08-31</td></tr><tr><td>6</td><td>6</td><td>Jeffery</td><td>Gan</td><td>NON_ACTIVE</td><td>2024-09-22</td></tr><tr><td>7</td><td>7</td><td>Ganesh</td><td>Kumar</td><td>ACTIVE</td><td>2025-10-27</td></tr><tr><td>8</td><td>8</td><td>Wyness</td><td>Ong</td><td>ACTIVE</td><td>2025-08-17</td></tr><tr><td>9</td><td>9</td><td>Ibrahim</td><td>Ali</td><td>NON_ACTIVE</td><td>2024-10-23</td></tr><tr><td>10</td><td>10</td><td>Jasmine</td><td>Lee</td><td>ACTIVE</td><td>2025-09-10</td></tr></table>	student_id	room_id	fname	lname	status	checkin_date	1	1	Ahmad	Mohammed	ACTIVE	2025-08-02	2	2	Yee Wen	Lau	ACTIVE	2025-09-01	3	3	Cheryl	Cheong	ACTIVE	2025-08-15	4	4	Dora	Poh	ACTIVE	2025-09-05	5	5	Emily	Liew	ACTIVE	2025-08-31	6	6	Jeffery	Gan	NON_ACTIVE	2024-09-22	7	7	Ganesh	Kumar	ACTIVE	2025-10-27	8	8	Wyness	Ong	ACTIVE	2025-08-17	9	9	Ibrahim	Ali	NON_ACTIVE	2024-10-23	10	10	Jasmine	Lee	ACTIVE	2025-09-10																																							
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	<pre>INSERT INTO maintenance (room_id, issue_desc, severity, status, reported_on, resolved_on) VALUES (1, 'Water leakage in bathroom', 'MEDIUM', 'OPEN', '2025-10-10', NULL), (2, 'Broken cabinet', 'LOW', 'RESOLVED', '2025-09-15', '2025-09-16'), (5, 'Aircond not cold', 'HIGH', 'OPEN', '2025-11-01', NULL), (10, 'Window latch broken', 'LOW', 'OPEN', '2025-11-03', NULL), (7, 'Clogged toilet', 'HIGH', 'RESOLVED', '2025-10-05', '2025-10-05'), (4, 'Faulty wiring for study lamp', 'MEDIUM', 'OPEN', '2025-10-28', NULL), (2, 'Bed frame cracked', 'MEDIUM', 'RESOLVED', '2025-09-20', '2025-09-22'), (8, 'Wifi router not working', 'LOW', 'OPEN', '2025-11-05', NULL), (1, 'Shower heater broken', 'MEDIUM', 'RESOLVED', '2025-05-01', '2025-05-04'), (3, 'Door lock jammed', 'MEDIUM', 'RESOLVED', '2025-06-15', '2025-06-16'), (3, 'Lamp not function', 'LOW', 'OPEN', '2025-06-15', NULL), (5, 'Window cannot open', 'LOW', 'OPEN', '2025-06-15', NULL), (4, 'Bed frame cracked', 'MEDIUM', 'OPEN', '2025-06-15', NULL), (6, 'Wifi router not working', 'LOW', 'OPEN', '2025-06-15', NULL);</pre> Insert 14 data into maintenance table. Based on the room_id, we can know the issue occurs, the severity, the status, the reported	<table><tr><th>maint_id</th><th>room_id</th><th>issue_desc</th><th>severity</th><th>status</th><th>reported_on</th><th>resolved_on</th></tr><tr><td>1</td><td>1</td><td>Water leakage in bathroom</td><td>MEDIUM</td><td>OPEN</td><td>2025-10-10</td><td>NULL</td></tr><tr><td>2</td><td>2</td><td>Broken cabinet</td><td>LOW</td><td>RESOLVED</td><td>2025-09-15</td><td>2025-09-16</td></tr><tr><td>3</td><td>5</td><td>Aircond not cold</td><td>HIGH</td><td>OPEN</td><td>2025-11-01</td><td>NULL</td></tr><tr><td>4</td><td>10</td><td>Window latch broken</td><td>LOW</td><td>OPEN</td><td>2025-11-03</td><td>NULL</td></tr><tr><td>5</td><td>7</td><td>Clogged toilet</td><td>HIGH</td><td>RESOLVED</td><td>2025-10-05</td><td>2025-10-05</td></tr><tr><td>6</td><td>4</td><td>Faulty wiring for study lamp</td><td>MEDIUM</td><td>OPEN</td><td>2025-10-28</td><td>NULL</td></tr><tr><td>7</td><td>2</td><td>Bed frame cracked</td><td>MEDIUM</td><td>RESOLVED</td><td>2025-09-20</td><td>2025-09-22</td></tr><tr><td>8</td><td>8</td><td>Wifi router not working</td><td>LOW</td><td>OPEN</td><td>2025-11-05</td><td>NULL</td></tr><tr><td>9</td><td>1</td><td>Shower heater broken</td><td>MEDIUM</td><td>RESOLVED</td><td>2025-05-01</td><td>2025-05-04</td></tr><tr><td>10</td><td>3</td><td>Door lock jammed</td><td>MEDIUM</td><td>RESOLVED</td><td>2025-06-15</td><td>2025-06-16</td></tr><tr><td>11</td><td>3</td><td>lamp not function</td><td>LOW</td><td>OPEN</td><td>2025-06-15</td><td>NULL</td></tr><tr><td>12</td><td>5</td><td>Window cannot open</td><td>LOW</td><td>OPEN</td><td>2025-06-15</td><td>NULL</td></tr><tr><td>13</td><td>4</td><td>Bed frame cracked</td><td>MEDIUM</td><td>OPEN</td><td>2025-06-15</td><td>NULL</td></tr><tr><td>14</td><td>6</td><td>Wifi router not working</td><td>LOW</td><td>OPEN</td><td>2025-06-15</td><td>NULL</td></tr></table>	maint_id	room_id	issue_desc	severity	status	reported_on	resolved_on	1	1	Water leakage in bathroom	MEDIUM	OPEN	2025-10-10	NULL	2	2	Broken cabinet	LOW	RESOLVED	2025-09-15	2025-09-16	3	5	Aircond not cold	HIGH	OPEN	2025-11-01	NULL	4	10	Window latch broken	LOW	OPEN	2025-11-03	NULL	5	7	Clogged toilet	HIGH	RESOLVED	2025-10-05	2025-10-05	6	4	Faulty wiring for study lamp	MEDIUM	OPEN	2025-10-28	NULL	7	2	Bed frame cracked	MEDIUM	RESOLVED	2025-09-20	2025-09-22	8	8	Wifi router not working	LOW	OPEN	2025-11-05	NULL	9	1	Shower heater broken	MEDIUM	RESOLVED	2025-05-01	2025-05-04	10	3	Door lock jammed	MEDIUM	RESOLVED	2025-06-15	2025-06-16	11	3	lamp not function	LOW	OPEN	2025-06-15	NULL	12	5	Window cannot open	LOW	OPEN	2025-06-15	NULL	13	4	Bed frame cracked	MEDIUM	OPEN	2025-06-15	NULL	14	6	Wifi router not working	LOW	OPEN	2025-06-15	NULL
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dates and the resolved date of the maintenance. The date of maintenance could be null when maintenance have not complete.

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INSERT INTO payments (student_id, amount, paid_on, method, note)
VALUES
-- August Payments
(1, 300.00, '2025-08-03', 'FPX', 'August Rent'), -- Ahmad (Checked in 8/2)
(3, 400.00, '2025-08-15', 'CARD', 'August Rent'), -- Cheryl (Checked in 8/15)
(8, 1200.00, '2025-08-18', 'TNG', 'August Rent'), -- Wyness (Checked in 8/17)
(5, 500.00, '2025-08-31', 'FPX', 'August Rent'), -- Emily (Checked in 8/31)

-- September Payments (for students checked in by Sep)
(1, 300.00, '2025-09-01', 'FPX', 'September Rent'), -- Ahmad (Checked in agian)
(2, 400.00, '2025-09-02', 'CARD', 'September Rent'), -- Yee Wen (Checked in 9/1)
(10, 1400.00, '2025-09-10', 'TNG', 'September Rent'), -- Jasmine (Checked in 9/10)
(4, 500.00, '2025-09-08', 'FPX', 'September Rent'),

-- October Payments (for students checked in by Oct)
(1, 300.00, '2025-10-01', 'FPX', 'October Rent'), -- Ahmad
(7, 900.00, '2025-10-27', 'CASH', 'October Rent'); -- Ganesh (Checked in 10/27)
```

Insert 10 data into payments table. Based on the above tables, calculate the amount for rent of students with the payment date, method and notes.

</

3.

SELECT * FROM room_types

WHERE rent BETWEEN 400 AND 800;

Display only the rows with rent between 400 and 800 from room_types table.

SELECT * FROM students

WHERE fname LIKE 'A%';

Display only the rows that first name (fname) starts with A from students table.

SELECT * FROM payments

WHERE method IN ('FPX','CARD');

Display only the rows where the method column is with 'FPX' or 'CARD' from payments table.

SELECT * FROM payments

WHERE (amount BETWEEN 300 AND 800 AND NOT method = 'TNG') OR method = 'CARD';

Display only the amount where it is between 300 and 800 but not the method with 'TNG'. However, it can be method with 'CARD'.

type_id	type_name	rent	deposit	capacity
2	Double Standard	400.00	200.00	2
3	Single Deluxe	400.00	200.00	1
4	Double Deluxe	500.00	250.00	2
5	Single Premium	500.00	250.00	1
6	Double Premium	600.00	300.00	2
NULL	NULL	NULL	NULL	NULL

student_id	room_id	fname	lname	status	checkin_date	email
1	1	Ahmad	Mohammed	ACTIVE	2025-08-02	ahmadm@graduate.utm.my
NULL	NULL	NULL	NULL	NULL	NULL	NULL

payment_id	student_id	amount	paid_on	method	note
11	1	300.00	2025-08-03	FPX	August Rent
12	3	400.00	2025-08-15	CARD	August Rent
14	5	500.00	2025-08-31	FPX	August Rent
15	1	300.00	2025-09-01	FPX	September Rent
16	2	400.00	2025-09-02	CARD	September Rent
18	4	500.00	2025-09-08	FPX	September Rent
19	1	300.00	2025-10-01	FPX	October Rent
NULL	NULL	NULL	NULL	NULL	NULL

payment_id	student_id	amount	paid_on	method	note
11	1	300.00	2025-08-03	FPX	August Rent
12	3	400.00	2025-08-15	CARD	August Rent
14	5	500.00	2025-08-31	FPX	August Rent
15	1	300.00	2025-09-01	FPX	September Rent
16	2	400.00	2025-09-02	CARD	September Rent
18	4	500.00	2025-09-08	FPX	September Rent
19	1	300.00	2025-10-01	FPX	October Rent
NULL	NULL	NULL	NULL	NULL	NULL

4.

a. Count()

SELECT COUNT(*) FROM students

WHERE status = 'ACTIVE'

Count the number of students where the status is stated 'ACTIVE' from students table.

b. SUM()

SELECT SUM(amount) FROM payments

WHERE method = 'FPX';

Calculate the total amount of the payments table where the method is 'FPX'.

a.

COUNT(*)
8

b.

SUM(amount)
1900.00

a. CONCAT()

```
SELECT CONCAT(lname, ' ', fname) AS Full_Name  
FROM students;
```

We join the last name (lname) and the first name (fname) from students table into a new column names 'Full_Name' using CONCAT string function.

b. UPPER()

```
SELECT UPPER(lname) AS UpperLastName  
FROM students
```

We use UPPER function to make the last name (lname) in students table into uppercase in a new column named 'UpperLastName'.

Full_Name
Mohammed Ahmad
Lau Yee Wen
Cheong Cheryl
Poh Dora
Liew Emily
Gan Jeffery
Kumar Ganesh
Ong Wyness
Ali Ibrahim
Lee Jasmine

a.

UpperLastName
MOHAMMED
LAU
CHEONG
POH
LIEW
GAN
KUMAR
ONG
ALI
LEE

b.

Task 3 – Reporting and Aggregation (CLO2)

	Code	Output																																																																																								
1.	<pre>CREATE OR REPLACE VIEW v_room_status AS SELECT r.room_no, rt.type_name, rt.rent, r.floor_no, rt.capacity, (SELECT COUNT(*) FROM students s WHERE s.room_id = r.room_id AND s.status = 'ACTIVE') AS n_occupants, (SELECT COUNT(*) FROM maintenance m WHERE m.room_id = r.room_id AND m.status = 'OPEN') AS pending_issues, CASE WHEN (SELECT COUNT(*) FROM students s WHERE s.room_id = r.room_id AND s.status = 'ACTIVE') = 0 THEN 'TRUE' ELSE 'FALSE' END AS is_vacant FROM rooms r JOIN room_types rt ON r.type_id = rt.type_id;</pre> Create a view named v_room_status with the columns room_no, type_name, rent, floor_no, capacity, n_occupants, pending_issues, and is_vacant. The value n_occupants represents the count of students with status = 'ACTIVE' in each room. The value pending_issues represents the count of maintenance records with status = 'OPEN' per room. The value is_vacant is TRUE when n_occupants = 0, otherwise FALSE.	<table><tr><th>room_no</th><th>type_name</th><th>rent</th><th>floor_no</th><th>capacity</th><th>n_occupants</th><th>pending_issues</th><th>is_vacant</th></tr><tr><td>A101</td><td>Single Standard</td><td>300.00</td><td>1</td><td>1</td><td>1</td><td>1</td><td>FALSE</td></tr><tr><td>B111</td><td>Double Standard</td><td>400.00</td><td>1</td><td>2</td><td>1</td><td>0</td><td>FALSE</td></tr><tr><td>A201</td><td>Single Deluxe</td><td>400.00</td><td>2</td><td>1</td><td>1</td><td>0</td><td>FALSE</td></tr><tr><td>B211</td><td>Double Deluxe</td><td>500.00</td><td>2</td><td>2</td><td>1</td><td>1</td><td>FALSE</td></tr><tr><td>A301</td><td>Single Premium</td><td>500.00</td><td>3</td><td>1</td><td>1</td><td>1</td><td>FALSE</td></tr><tr><td>B311</td><td>Double Premium</td><td>600.00</td><td>3</td><td>2</td><td>0</td><td>0</td><td>TRUE</td></tr><tr><td>C401</td><td>Family Standard (Small)</td><td>900.00</td><td>4</td><td>4</td><td>1</td><td>0</td><td>FALSE</td></tr><tr><td>C411</td><td>Family Standard (Large)</td><td>1200.00</td><td>4</td><td>6</td><td>1</td><td>1</td><td>FALSE</td></tr><tr><td>C501</td><td>Family Deluxe (Small)</td><td>1100.00</td><td>5</td><td>1</td><td>0</td><td>0</td><td>TRUE</td></tr><tr><td>C512</td><td>Family Deluxe (Large)</td><td>1400.00</td><td>5</td><td>2</td><td>1</td><td>1</td><td>FALSE</td></tr></table>	room_no	type_name	rent	floor_no	capacity	n_occupants	pending_issues	is_vacant	A101	Single Standard	300.00	1	1	1	1	FALSE	B111	Double Standard	400.00	1	2	1	0	FALSE	A201	Single Deluxe	400.00	2	1	1	0	FALSE	B211	Double Deluxe	500.00	2	2	1	1	FALSE	A301	Single Premium	500.00	3	1	1	1	FALSE	B311	Double Premium	600.00	3	2	0	0	TRUE	C401	Family Standard (Small)	900.00	4	4	1	0	FALSE	C411	Family Standard (Large)	1200.00	4	6	1	1	FALSE	C501	Family Deluxe (Small)	1100.00	5	1	0	0	TRUE	C512	Family Deluxe (Large)	1400.00	5	2	1	1	FALSE
room_no	type_name	rent	floor_no	capacity	n_occupants	pending_issues	is_vacant																																																																																			
A101	Single Standard	300.00	1	1	1	1	FALSE																																																																																			
B111	Double Standard	400.00	1	2	1	0	FALSE																																																																																			
A201	Single Deluxe	400.00	2	1	1	0	FALSE																																																																																			
B211	Double Deluxe	500.00	2	2	1	1	FALSE																																																																																			
A301	Single Premium	500.00	3	1	1	1	FALSE																																																																																			
B311	Double Premium	600.00	3	2	0	0	TRUE																																																																																			
C401	Family Standard (Small)	900.00	4	4	1	0	FALSE																																																																																			
C411	Family Standard (Large)	1200.00	4	6	1	1	FALSE																																																																																			
C501	Family Deluxe (Small)	1100.00	5	1	0	0	TRUE																																																																																			
C512	Family Deluxe (Large)	1400.00	5	2	1	1	FALSE																																																																																			
2.	<pre>SELECT rt.type_name, COUNT(s.student_id) AS Total_Student FROM room_types rt JOIN rooms r ON rt.type_id = r.type_id LEFT JOIN students s ON r.room_id = s.room_id GROUP BY type_name;</pre> Retrieve the total number of students grouped by room type.	<table><tr><th>type_name</th><th>Total_Student</th></tr><tr><td>Single Standard</td><td>1</td></tr><tr><td>Double Standard</td><td>1</td></tr><tr><td>Single Deluxe</td><td>1</td></tr><tr><td>Double Deluxe</td><td>1</td></tr><tr><td>Single Premium</td><td>1</td></tr><tr><td>Double Premium</td><td>1</td></tr><tr><td>Family Standard (Small)</td><td>1</td></tr><tr><td>Family Standard (Large)</td><td>1</td></tr><tr><td>Family Deluxe (Small)</td><td>1</td></tr><tr><td>Family Deluxe (Large)</td><td>1</td></tr></table>	type_name	Total_Student	Single Standard	1	Double Standard	1	Single Deluxe	1	Double Deluxe	1	Single Premium	1	Double Premium	1	Family Standard (Small)	1	Family Standard (Large)	1	Family Deluxe (Small)	1	Family Deluxe (Large)	1																																																																		
type_name	Total_Student																																																																																									
Single Standard	1																																																																																									
Double Standard	1																																																																																									
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Family Deluxe (Small)	1																																																																																									
Family Deluxe (Large)	1																																																																																									

```
SELECT
    type_name,
    ROUND(AVG(rent),2) AS Avg_Rent,
    ROUND(SUM(deposit),2) AS Total_Deposit
FROM room_types
GROUP BY type_name;
```

Retrieve the average rent and total deposit per room type.

The values are rounded to 2 decimal places using ROUND()

```
SELECT
    YEAR(paid_on) AS Year,
    MONTH(paid_on) AS Month,
    SUM(amount) AS Total_Amount
FROM payments
GROUP BY YEAR(paid_on) , MONTH(paid_on)
ORDER BY Year, Month;
```

Retrieve total payment amounts grouped by year and month.

```
SELECT
    r.floor_no,
    COUNT(m.maint_id) AS open_issues
FROM maintenance m
JOIN rooms r
    ON m.room_id = r.room_id
WHERE m.status = 'OPEN'
GROUP BY r.floor_no
HAVING COUNT(m.maint_id) > 2;
```

Retrieve the count of OPEN maintenance issues per floor.

Only shows floors with more than 2 issues using HAVING COUNT > 2.

```
SELECT
    UPPER(rt.type_name) AS ROOM_TYPE,
    CONCAT('Floor ', r.floor_no, ', Room ', r.room_no) AS FULL_LOCATION,
    ROUND(rt.rent, 2) AS RENT_AMOUNT,
    CASE
        WHEN rt.rent < 500 THEN 'LOW'
        WHEN rt.rent BETWEEN 500 AND 800 THEN 'MEDIUM'
        ELSE 'HIGH'
    END AS RENT_CATEGORY
FROM rooms r
JOIN room_types rt ON r.type_id = rt.type_id
ORDER BY rt.rent, r.floor_no;
```

We make the room type to uppercase, join the floor number and room number together in a

	type_name	Avg_Rent	Total_Deposit
▶	Single Standard	300.00	150.00
	Double Standard	400.00	200.00
	Single Deluxe	400.00	200.00
	Double Deluxe	500.00	250.00
	Single Premium	500.00	250.00
	Double Premium	600.00	300.00
	Family Standard (Small)	900.00	500.00
	Family Standard (Large)	1200.00	600.00
	Family Deluxe (Small)	1100.00	300.00
	Family Deluxe (Large)	1400.00	250.00

	Year	Month	Total_Amount
▶	2025	8	2400.00
	2025	9	2600.00
	2025	10	1200.00

	floor_no	open_issues
	3	3
	2	3

	ROOM_TYPE	FULL_LOCATION	RENT_AMOUNT	RENT_CATEGORY
▶	SINGLE STANDARD	Floor 1, Room A101	300.00	LOW
	DOUBLE STANDARD	Floor 1, Room B111	400.00	LOW
	SINGLE DELUXE	Floor 2, Room A201	400.00	LOW
	DOUBLE DELUXE	Floor 2, Room B211	500.00	MEDIUM
	SINGLE PREMIUM	Floor 3, Room A301	500.00	MEDIUM
	DOUBLE PREMIUM	Floor 3, Room B311	600.00	MEDIUM
	FAMILY STANDARD (SMALL)	Floor 4, Room C401	900.00	HIGH
	FAMILY DELUXE (SMALL)	Floor 5, Room C501	1100.00	HIGH
	FAMILY STANDARD (LARGE)	Floor 4, Room C411	1200.00	HIGH
	FAMILY DELUXE (LARGE)	Floor 5, Room C512	1400.00	HIGH

	column named 'FULL_LOCATION', round up the rental amount into 2 decimal places and show the price category. All in ascending order.	
--	---	--